Redundancy Repartitions Capacity and Enhances Recall in Visual Working Memory: The Sample-Size Model of Neural Activation

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Abstract

Visual working memory (VWM) has a limited capacity of around four items, yet recall is also improved by structure in the memory array. We reconcile these views by testing how color redundancy – a low-level structural regularity – alters recall precision in a continuous color-reproduction task using a between-subjects (21 participants) and small-N within-subjects design (5 participants). We model precision by combining the attention-weighted sample-size model of Smith et al. (2018) with Oberauer's (2023) signal discrimination model. This sample-size model of neural activation describes baseline predictions for how capacity limits and recall sensitivity should theoretically improve with redundancy, and can be used as a diagnostic process model of VWM and attention. Our model shows that array regularities enhance overall item capacity by storing redundant items as unitary objects and that redundant items often benefit from probability summation. Our model also identifies nuanced individual attentional strategies: trade-offs in redundant and non-redundant item recall, redundant-item neural summation, and Gestalt-like encoding strategies across redundant and non-redundant items. These individual differences would be masked by data aggregation. Our results motivate the use of individual-level modeling when theorizing about nuanced relationships, such as exists between attention and VWM.

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Conflicts of interest/Competing interests

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Availability of data and material

All data will be available at osf.io/pxskw upon publication of the manuscript.

Code availability

All experimental and analysis code will be available at osf.io/pxskw upon publication of the manuscript.

Authors' contributions

All authors contributed to the study conception, ideas for data analysis, manuscript writing and editing. Y.L. collected data. P.G. programmed the experiment. Y.L and P.G analyzed the data and share primary authorship.

Ethics approval

Ethics approval was received from The University of Melbourne Human Research Ethics Committee (approval 1646144.5).

Consent to participate

Each participant provided informed online consent at the start of each session in Experiment 1, and provided written consent at the start of the first testing session in Experiment 2.

Redundancy Repartitions Capacity and Enhances Recall in Visual Working Memory: The Sample-Size Model of Neural Activation

Visual working memory (VWM) is the roughly four-item, capacity-limited bottleneck in translating perception into action (Phillips, 1974; Luck & Vogel, 1997; W. Zhang & Luck, 2008). Researchers have proposed two conflicting accounts of this limit. One account posits that when an observer views a memory array – a set of features (colors, orientations) or objects (feature-conjunctions) – VWM stores a small set of minimally interacting, integrated objects. The total count of these objects determines recall performance (Luck & Vogel, 1997; Cowan, 2001; Vogel et al., 2001; W. Zhang & Luck, 2008; Luria et al., 2010), with some additional object-complexity constraints (e.g., Alvarez & Cavanagh, 2004). The other account holds that VWM representations remain sensitive to higher-order ensemble statistics. For example, recalled items benefit from probabilistic (Markov) structures within an array (Brady & Tenenbaum, 2013; Orhan & Jacobs, 2013; Brady & Alvarez, 2015) and may be biased toward the average array color (Brady & Alvarez, 2011).

This article investigates how item-capacity limits interact with memory array structure by manipulating redundancy. By repeating colors within a memory array and probing memory on a continuous color wheel, we test whether redundant colors yield precision and response time (RT) advantages over uniquely colored, non-redundant items within the same mixed array. We introduce the *attention-weighted sample-size model of neural activation* to test how redundancy alters VWM capacity allocation, and whether redundant items benefit by minimally interacting with, or by redistributing mnemonic resources away from their non-redundant counterparts. The fundamental goal of this paper is to provide a tightly constrained and theoretically informed account of how redundancy interacts with VWM.

Redundant Item Gains

Redundancy gains – faster or more accurate redundant-item responses – have long been a focus in the cognitive architecture literature (Raab, 1962; Miller, 1982),

particularly by Townsend and colleagues' Systems Factorial Technology (Townsend & Nozawa, 1995; Eidels et al., 2008; Townsend & Eidels, 2011; Eidels et al., 2015; D. R. Little et al., 2015; D. Little et al., 2017; Yang et al., 2018; D. R. Little et al., 2019; Garrett et al., 2019; Howard et al., 2021; H. Zhang, Garrett, Lin, et al., 2023; H. Zhang, Garrett, Houpt, et al., 2023; Fong et al., 2024). Redundancy gains may arise because the visual system pools redundant information into an integral representation (e.g., first-order 'coactivation' in the cognitive architecture literature or 'neural summation' in the psychophysics literature; Townsend & Nozawa, 1995 and Graham 1977, respectively); because redundancy creates a structural Gestalt that may be stored as a singular object in VWM (a second-order post-encoding process; Shaw, 1980); because independent memory copies race in parallel during encoding and retrieval (statistical facilitation or second-order probability summation; Raab, 1962; Bower, 1967; Thornton & Gilden, 2007); or because the brain redirects attention or memory resources away from non-redundant elements to facilitate faster and more precise redundant item recall. Under a limited capacity, parallel (simultaneous) encoding strategy, these three mechanisms predict on average, faster and more accurate redundant item retrieval in a mixed item array – an array containing both redundant and non-redundant elements. However, these same mechanisms predict very different outcomes when recalling non-redundant elements.

Non-Redundant Item Effects

Gestalt Encoding Benefits. Substantial evidence suggests that the unit of VWM encoding is the perceptual object, rather than isolated features (e.g., Luck & Vogel, 1997; Vogel et al., 2001). The simplest version of 'Gestalt encoding' holds that the visual system encodes a set of redundant elements unitarily, as a single perceptual object or memory representation (Figure 1B). If redundancy leads to unitary color encoding, then working memory performance – typically limited to around four elements – should scale with the number of distinct perceptual objects (colors) rather than the number of physically separate items. This can be thought of as improving the

signal-to-noise ratio (Green & Swets, 1966) via information compression (à la rate distortion theory; Berger, 1971), or as reducing inter-item confusability (Kinchla, 1974). Ensemble or Gestalt encoding does not necessitate neural summation (Whitney & Yamanashi Leib, 2018) and does not selectively benefit redundant items. Rather, in a mixed array, Gestalt encoding reduces the effective item unit N – that which describes memory limits – from the *number of individual items* to the *number of unique item colors*, benefiting both redundant and non-redundant item recall alike (Figure 1B).

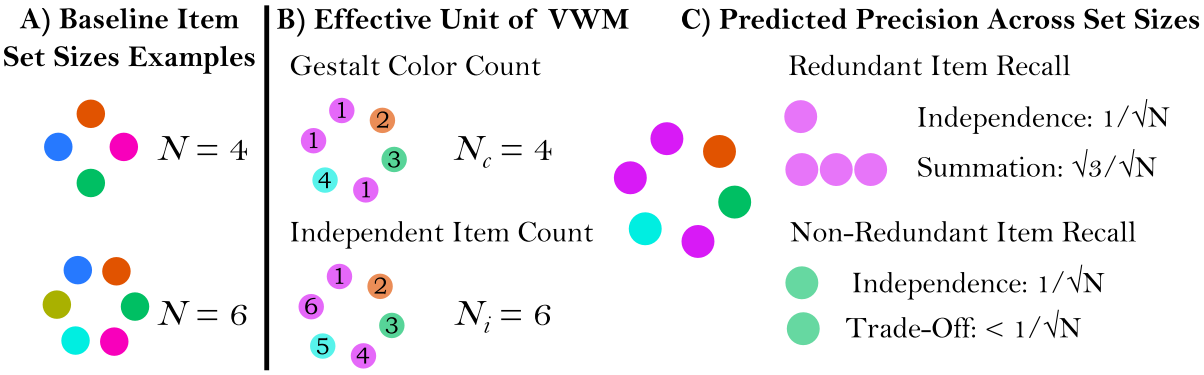


Figure 1. A) Illustrates unique item sets. Recall precision for these baseline sets are directly comparable to performance in the mixed redundant-item arrays (Panel B) under the assumption of Gestalt encoding ($N_c = 4$) or independent item counts ($N_i = 6$). C) Shows how recall precision may improve for redundant items (top) under the assumption of probability summation or information integration per the sample-size model (see how we define probability summation processes in text). Non-redundant item performance (bottom) may be independent from or trade-off with redundancy gains.

Independent Redundancy Gains. Three broad theoretical classes explain why redundancy gains may minimally impact non-redundant item recall: Parallel statistical facilitation (Raab, 1962) and probability summation (Green & Swets, 1966); neural summation (Graham, 1977) or coactivation (Townsend & Nozawa, 1995); and swap errors (Bays et al., 2009, 2011).

Statistical facilitation and probability summation respectively describe how faster and more accurate responding can arise when independent 'copies' (e.g., Bower, 1967) race in parallel during encoding and retrieval. Parallel independent races improve the likelihood of a fast and precise redundant item retrieval from memory, without affecting non-redundant item recall. Independence in this context describes the encoding or retrieval of information into VWM units, which could encompass a singular feature (one

color) or an entire object (a set of redundant colors). We will use the term 'probability summation' throughout to refer to the statistical benefits obtained by aggregating multiple noisy sources of information, but we do not prescribe how this benefit accrues. Our definition of probability summation therefore encompasses benefits that arise from aggregating the outputs of a set of independent, noisy detectors (Quick Jr, 1974) or by summing a set of noisy strength variables as in optimum likelihood signal detection models Green & Swets (1966); the predictions of which are almost identical (Nolte & Jaarsma, 1967). It likewise covers temporal information integration as observed in multiple-look designs (Lilburn & Smith, 2020; Swets et al., 1959). These second-order summation processes (Shaw, 1980) – statistical facilitation and probability summation – provide the baseline benefit predicted by independent parallel structures, beyond which would require redundant items to be advantaged by attentional trade-offs or neural summation.

Neural summation and coactivation describe the super-additive performance of a parallel processing system. These processes predict performance greater than the sum of their independent components (a first order summation process), exceeding the predictions of probability summation (Miller, 1982). Under a neural summation account, redundant item recall performance may benefit in a mixed array without negatively impacting the performance of non-redundant elements (for an example of how layered parallel (in)dependent perceptual structures may co-exist, see the CODE theory framework; Logan, 1996).

Finally, swap errors – where observers incorrectly report the location of a non-target item in place of a probed target (Bays et al., 2009, 2011) – may selectively benefit redundant item recall. If we assume that swap errors reflect location binding errors, then when cued to report a redundant item, a swap error may reproduce a different yet identically colored item from memory: a correct response. By contrast, non-redundant swap errors necessarily result in a non-target color: an incorrect response. However, if swap errors are not location bound and instead occur between unique colors stored in memory, then under a Gestalt framework, these memory swaps

would equally affect both redundant and non-redundant responses as each unique color only occupies one 'unit' of VWM.

Redundancy Trade-Offs. Redundant item benefits may come at a cost to non-redundant item memory performance due to attentional biases. For example, exogenous biases, such as attention capture (Chen et al., 2019) – where redundant items engage visual attention before non-redundant items in a mixed array – may bias encoding and improve redundant item recall, but at a cost to the encoding and performance of non-redundant items. Similarly, attentional strategies where participants choose to actively prioritize redundant item information during encoding or retention, may improve redundant item recall, but at a cost to non-redundant recall performance.

Finally, we note that redundancy gains may boost non-redundant item performance by freeing memory resources. These 'spill-over' benefits have been observed in change detection tasks (Morey et al., 2015; Morey, 2019) but have not been observed in probed recall tasks (e.g., Ramzaoui et al., 2025). As we will be using a probed recall task, we do not expect participants to display spill-over benefits.

Necessary model predictions. Establishing how redundancy affects non-redundant item recall requires a model that can evaluate the following predictions: First, whether the effective unit of VWM is determined by the number of items or number of colors in the array (H_1 the 'Gestalt encoding' hypothesis). Second, whether redundant recall performance is best accounted for by independent neural firing (no redundancy gain) or by our broad definition of probability summation (H_2 the 'probability summation' hypothesis). Further, the model must disambiguate whether redundancy performance exceeds probability summation predictions by reallocating non-redundant item attention (the 'attention reallocation' hypothesis) or by the super-additive coactivation of redundant elements in the array (the 'neural summation' hypothesis). We will call these collective hypotheses H_{2a} as they will trade off in our model. Consequently, we treat evidence that selective attention benefits redundant items and costs non-redundant items, as evidence against neural summation. Although the two processes could occur in tandem, they typically reflect dichotomous process

efficiencies: super-additivity and limited capacity (Townsend & Nozawa, 1995). To formalize the quantitative predictions of our proposed mechanisms and how they scale with set size, we next consider the sample-size model.

The sample-size model

The sample-size model predicts that, for a given set size N, memory precision – circular standard deviation in continuous tasks or d' in two-choice tasks – equals single-item precision divided by the $\sqrt{N}$. This $1/\sqrt{N}$ law recurs throughout psychophysics and vision research (Rose, 1948; Swets et al., 1959; Taylor et al., 1967; Lindsay et al., 1968; Shaw, 1980, 1982, 1984; Palmer, 1990; Bonnel & Miller, 1994; Bonnel & Hafter, 1998) and appears in VWM studies (e.g., Palmer, 1990; Wilken & Ma, 2004; Van den Berg et al., 2012; Bays, 2014; Van den Berg et al., 2014; Schneegans et al., 2020). Sewell et al. (2014) formalized this relationship as a capacity metric, noting that recall sensitivity declined following a $1/\sqrt{N}$ relationship, regardless of whether memory array items were presented simultaneously (encoding limited) or sequentially (capacity limited).

Extending on this finding, Smith et al. (2016, 2018) proposed the *attention-weighted sample-size model*. In this model, items that attract more attention receive a larger resource share, producing a power-law with exponent β: $\beta = -0.5$ matches the sample-size model, $\beta < -0.5$ represents attentionally demanding stimuli (as theorized in Smith et al., 2018; and seen in Bays & Husain, 2008 and Van den Berg et al., 2014), and $\beta > -0.5$ represent attentional improvements relative to sample-size predictions (e.g., benefits arising from probability summation, neural summation, or attentional trade-offs).

Subsequent work has supported and extended the sample-size model (Corbett & Smith, 2017; Smith et al., 2018; Lilburn & Smith, 2020). Notably, Lilburn and Smith (2020) implemented a two-interval stimulus presentation paradigm and showed a $\sqrt{2}$ improvement in d' sensitivity, as predicted by probability summation. We refer to this as a $\sqrt{m}/\sqrt{N}$ sensitivity gain, where m is the number of repeated presentations or

redundant elements. Fong et al. (2024) then showed that multiple intervals and redundancy have similar effects on sample-size model predictions. Moreover, their four-item change detection task showed convergent array-size capacity limitations using both an RT measure – the capacity coefficient (Townsend & Nozawa, 1995) – and the sample-size model when applied to d'.

Most recently, the sample-size model has been embedded into neural population coding frameworks. Qian et al. (2025) showed that neural-activation amplitudes decrease as array sizes increase, following a sample-size rule. Using the same data, Smith et al. (2025) demonstrated through circular diffusion modeling that the latent memory component of VWM obeys a $1/\sqrt{N}$ law while attention modulates departures ($\beta > -.5$ or $\beta < -.5$). These findings support the view that VWM capacity reflects a fixed pool of noisy samples apportioned across items, with attention influencing how evenly those samples are distributed within the memory array.

In summary, the sample-size model affords a measure of memory capacity, N, a baseline prediction of capacity performance assuming item independence, $1/\sqrt{N}$, and a baseline prediction assuming probability summation, $\sqrt{m}/\sqrt{N}$. To embed these predictions in an established VWM framework – one used to predict the distribution of recall errors as a function of set size – we turn to Oberauer's (2023) Signal Discrimination Model (SDM).

The Signal Discrimination Model (SDM)

In the SDM, each item enters memory with its strength specified by a circular tuning function over a population of feature-tuned neurons. Following (2023), we assume the tuning function has the form of a von Mises distribution – the circular analogue of a Gaussian (as proposed by Bays, 2014; Oberauer, 2023). Independent Gumbel noise perturbs each neuron's activation, and the reported item is taken to be the maximally responding member of this noisy ensemble (a "Gumbel-max" or random-utility argmax choice rule).

Oberauer showed that the predictions of the SDM are relatively insensitive to the

form of the noise function, but that the Gumbel noise model lends itself to a simple expression for the choice probabilities. Specifically, under these assumptions, *response probabilities* obey Luce's choice rule: the probability of choosing item i is proportional to its strength relative to the summed strengths of all items (e.g., Yellott, 1977; Balog et al., 2017). In other words, the Gumbel-Luce decision mechanism induces choice probabilities with the same divisive-normalization form as models of cortical population coding (Carandini & Heeger, 2012): an item's effective signal is its own strength divided by the sum of strengths across items. Importantly, this normalization arises at the level of response probabilities rather than as a biophysical operation on neural firing rates (see Qian et al. 2025, for tests of alternative tuning and noise functions within this framework).

Oberauer framed the SDM as a measurement model that, when paired with an interference account (Oberauer & Lin, 2017), provides a full process model. Building on work explicitly linking power-law (sample-size-like) scaling to neural population codes (e.g., Bays, 2014; Smith et al., 2016; Schneegans et al., 2020), we contend that the simpler sample-size model may be applied in the SDM framework to similar effect. We refer to this implementation as the *sample-size model of neural activation*.

The (attention-weighted) sample-size model of neural activation

The SDM captures declining recall precision across set sizes through a declining memory-strength (amplitude, A) parameter and fixed neural tuning-width parameter (κ). This decline in amplitude can be described by a power-law. Work by Qian et al. (n.d.) has shown that SDM model fits conditioned on a declining amplitude power-law (rather than a declining κ) best account for data under a maximum-output decoding rule.

Figure 2A illustrates declining response precision across array sizes. Distributions share a single amplitude (A_1) and concentration (κ) parameter, and amplitude follows a power-law exponent (β). Independent neural firing – our assumed baseline for unique-item color arrays – follows the sample-size model predictions, where amplitude

at N items matches $A_1 N^{-0.5}$. *Across* trials, activation declines as N increases, with sample-size predictions matching the central black curve in Figure 2A. However, *within* a trial, activation is equally weighted per item, as in Figure 2B.

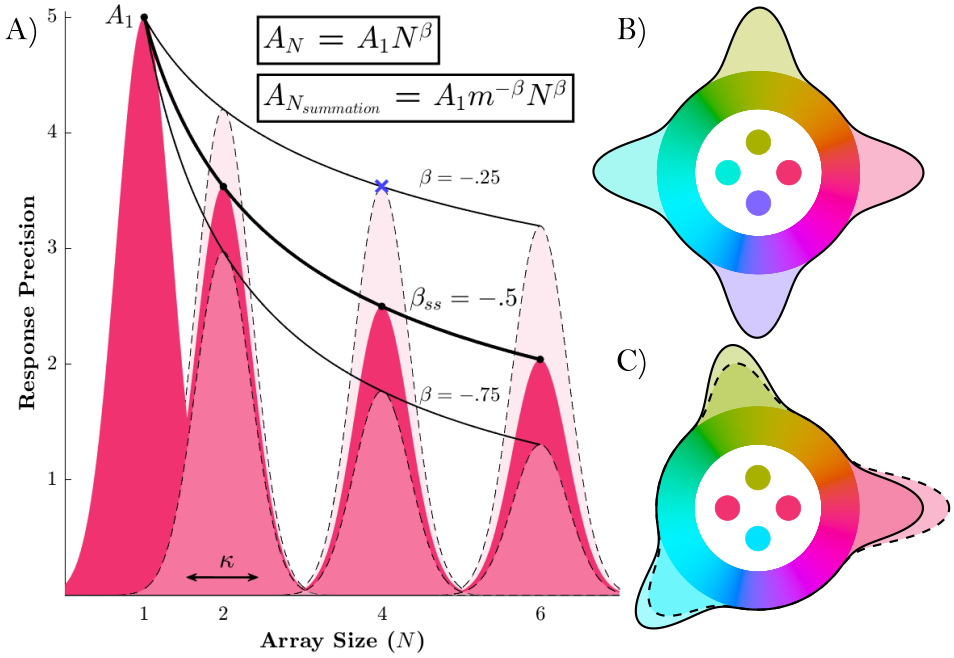


Figure 2. A) Decreasing von Mises distributions of response precision for array sizes 1, 2, 4, & 6, described by a single neural amplitude (A_1) parameter with fixed kappa (κ). Sample-size model predictions match β_{SS} = -0.5; weighted attention gains (β = -0.25) and costs (β = -0.75) are illustrated by modified power-laws. The blue 'X' illustrates predicted redundancy gains for a four-item, two-color condition (N = 4, m = 2, Panel C) assuming neural summation. B) Illustration of equal neural activations for four unique items on a color wheel. C) Solid-line illustrates sample-size/independent amplitudes; dashed line illustrates improved amplitude ($\beta >$ sample-size) for red-redundant items at the expense of unique item amplitudes ($\beta <$ sample-size; the attentional reallocation hypothesis).

The sample-size model conditions recall sensitivity – amplitude in the SDM framework – on a given set size, N. If redundant features function as a unitary object, this reduces the 'effective unit of VWM' from the total number of items, N_i, to the unique color count, N_c (Figure 1B). Model comparison testing N_i vs N_c sample-size relationships provide a direct test of the Gestalt encoding hypothesis.

As shown by Lilburn and Smith (2020) and Fong et al. (2024), probability summation may boost redundant item sensitivity – amplitude under the SDM framework. Assuming probability summation, the baseline sample-size model predicts a

$A_1\sqrt{m}/\sqrt{N}$ improvement for redundant item amplitude, as shown by the blue 'X' in Figure 2A (where $m = 2, N = 4$). Without probability summation, this would present as an unexplained boost in amplitude (e.g., the $\beta = -.25$ line). Assessing model fits that assume independent neural firing vs probability summation for redundant elements provides a direct test for the probability summation hypothesis.

Under the sample-size framework, if we observe probability summation, then our $A_1\sqrt{m}/\sqrt{N}$ prediction becomes the new baseline prediction for redundant item activation. Therefore, observing redundant item amplitudes beyond this new baseline implies neural summation or attentional biases toward redundant item recall. If non-redundant amplitude follows the baseline sample-size prediction (or at minimum, match the predictions of an identically sized 'unique color' array; Figure 1A) then we may infer non-redundant elements have minimally interacted with redundant mixed array elements: a neural summation account. However, if non-redundant item amplitude is worse than predicted by a matched unique color set (e.g., $\beta = -.75$ in Figure 2A), then a redundancy trade-off has occurred: the attention reallocation hypothesis. These baseline predictions are made while assuming array elements are non-interacting (e.g., not forming shape or pattern Gestalts that may assist memory) as the baseline model projects from a single-item's activation. The consequences of violating this assumption will be taken up in the Discussion.

The current paper

This paper introduces the attention-weighted sample-size model of neural activation to provide a process-model account of redundancy gains and costs at the individual-subject level. We begin with a larger-N study in which we explore robust group level redundancy effects, before addressing our study hypotheses – H_1 Gestalt encoding and H_2 redundancy gains via probability summation and H_{2a} neural summation or attentional biases – through small-N model fits that combine the SDM with highly constrained sample-size model predictions. Across five participants, our findings support Gestalt encoding, and show mixed support for probability summation,

attentional trade-offs, and swap errors. Redundant item neural summation was observed in one participant.

Method

We present a single method section for Experiments 1 and 2, noting differences therein. Primary findings are reported in text. Supplementary findings are reported in the appendix. Data and code are available through the Open Science Framework *osf.io/pxskw*. The University of Melbourne Human Research Ethics Committee approved both experiments (approval 1646144.5). These studies were not preregistered.

Design and Procedure

Figure 3 illustrates our 4 (set-size: 1, 2, 4, 6) × 4 (colors: 1, 2, 4, 6) × 2 (cue-type: whether a redundant vs non-redundant item was cued for recall) factorial design. Experiment 1 was a two-session, within-subjects online design and included a 'complete redundancy' condition (bottom row) wherein all items were the same color (as in Thornton & Gilden, 2007). Participants were University of Melbourne students and were awarded 2 course credits for their time.

Experiment 2 omitted the complete-redundancy condition and employed a 10-session small-N, within-subjects design wherein each participant completed 4500 trials. This affords individual-level modeling wherein each participant serves as an independent replication of the model findings (Smith & Little, 2018). Experiment 2 employed lab volunteers and was completed in a darkened testing booth in the Vision and Attention Laboratory at the University of Melbourne.

Consenting participants provided demographics – age, gender, color-blind status – and performed seven practice trials. Trials began by clicking a "start" button centered on a dark-gray screen (RGB 20, 20, 20). A 200 ms white fixation cross appeared, followed by a 400 ms stimulus array containing one-to-six colored discs. At a distance of 100 cm, each disc subtended 0.97° visual angle. Discs were arranged equidistantly from one another on the imaginary circumference of a 4.7° diameter circle. We generated 360 stimulus colors in Matlab using CIE-LUV space – chosen for perceptual uniformity in

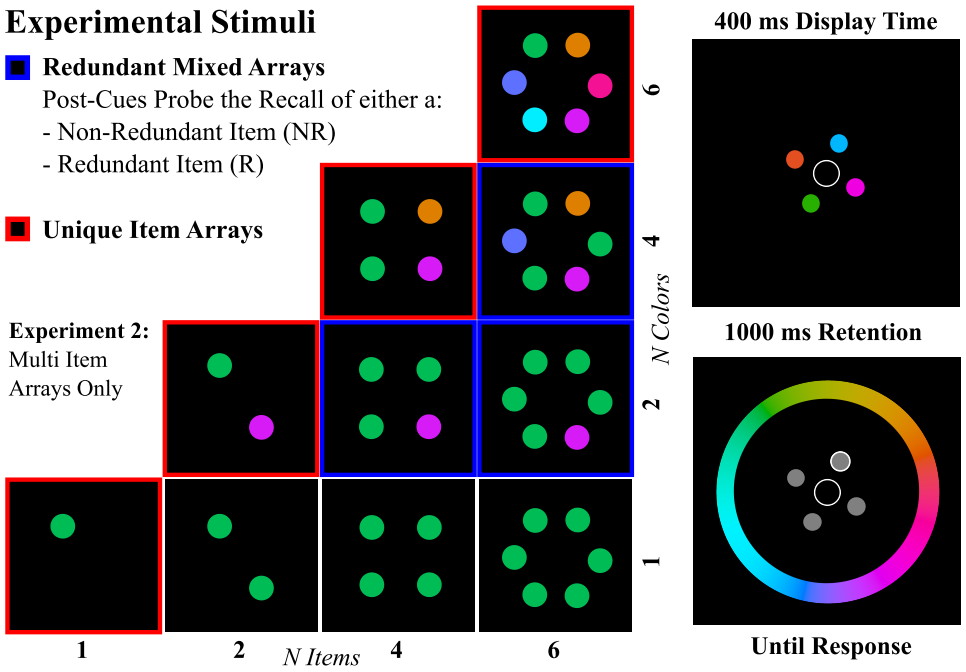


Figure 3. Left: Stimuli by item set-size and number of item colors. Redundant condition probes selected redundant and non-redundant items with equal probability. Complete redundancy (Experiment 1 only) and unique item conditions only cued redundant or non-redundant items, respectively. Right: Trial progression illustrating a stimulus (top), probe (bottom white outline) and surrounding color response wheel.

brightness – centered on the white point $L = 54$, $U = 0.0037$, and $V = -0.0275$, with a radius of 127, which we converted to sRGB for presentation in JSpsych (JavaScript), and were randomly selected for presentation with a minimum inter-color separation of $\pm 15°$.

Following the stimulus array, a 1000 ms blank retention interval preceded the response phase. Colored discs were replaced by gray patches (RGB 128, 128, 128); the target patch displayed a white border. Participants reported the target color by crossing the inner boundary of the color wheel (inner diameter = 24.38°, outer diameter = 26.45°). RT and angular error (precision) were recorded and feedback showing the distance between the response and the correct location was displayed.

To maintain a consistent starting position, a white circle (diameter = 1.26°) marked the screen center. If the cursor failed to start in this circle, left too quickly (< 300 ms), or lingered too long (> 3000 ms), we imposed a 4000 ms feedback penalty; otherwise trials ended with 1000 ms of feedback followed by a self-paced break screen displaying a graphical trial-by-trial precision summary.

In Experiment 1, participants viewed 810 randomly selected trials over two sessions comprising one, two, four or six items: 246 unique item trials, 120 complete redundancy trials, and 222 redundant-cued and 222 non-redundant cued trials (see Figure 3). In Experiment 2, each participant viewed 4500 trials over ten sessions comprising two, four or six items: 1500 unique item, 1500 redundant-cued, and 1500 non-redundant cued trials. Experiment 2 participants reported their recall strategies after their final testing session.

Participants

Thirty-four participants (M = 8, F = 26) aged 16 – 28 (M_{age} = 19, SD_{age} = 2) years participated in Experiment 1. Five Vision and Attention Lab volunteers (M = 4, F = 1) aged 21 – 41 (M_{age} = 28, SD_{age} = 8) years participated in Experiment 2. All participants reported normal or corrected-to-normal color vision.

Analyses

Mixture model screening

We excluded participants who displayed guessing patterns in the unique one- or two-item conditions. We fit a two-component mixture model (von Mises + uniform distributions) to the response-error data of each participant, fixing the mean (μ) at 0 and sharing one concentration parameter (κ) across unique-item arrays. The uniform-component weight was free to vary with array size and provided an estimate of 'guess-like' responses (Zhang & Luck, 2008). Participants were removed if the uniform proportion exceeded 10% in the one-item array or 20% in the two-item array (see Appendix A).

Descriptive means

To assess differences in Experiment 1 mean RT and circular standard deviation, frequentist (Greenhouse-Geisser corrected) and Bayesian repeated-measures ANOVAs were computed using the statistical software JAMOVI (2025). Bayesian ANOVAs

compared specified models (e.g., a set size effect across RT), against their null counterpart, reporting evidence as Bayes Factors away from the null (BF_{10}). Where multiple additive models were tested (e.g., set-size + cue-type), the highest posterior probability model was reported. Post-hoc tests were conducted using Westfall-Johnson-Utts corrections (Bayesian) and Tukey comparisons (Frequentist).

To visualize conditional differences in mean RT and circular SD in Experiment 1, we computed non-parametric 95% highest density intervals (HDIs). For individual subject error bars in Experiment 2, each participant's conditional data was sampled with replacement to create 5000 datasets, with a sliding window used to identify the narrowest region containing 95% of the bootstrap means – this HDI calculation prevents skewed distributions from misrepresenting the densest data region. For Experiment 1 group level error bars, intervals were estimated using a two-stage bootstrap: 5000 participant means were bootstrapped with a sliding HDI applied, with each participant sampled randomly with replacement using the aforementioned routine. This method yields conservative 95% HDIs across conditions and within individuals.

Recall Precision Modeling

Using data from Experiment 2, we implemented the sample size model of neural activation across nine experimental sub-conditions (omitting the complete redundancy condition; Table 1).

Table 1
Nine sub-conditions considered in the neural population coding model (Experiment 2).

N_i **Item Count**	N_c **Color Count**	**Description**	**Abbreviation**
2	2	Unique	2
4	4	Unique	4
4	2	Redundant Cued	4_{2R}
4	2	Non-Redundant Cued	4_{2NR}
6	6	Unique	6
6	2	Redundant Cued	6_{2R}
6	2	Non-Redundant Cued	6_{2NR}
6	4	Redundant Cued	6_{4R}
6	4	Non-Redundant Cued	6_{4NR}

We treated the neural noise at each detector as independent and identically

Gumbel-distributed. As discussed by Oberauer (2023), Yellott (1977) demonstrated that when noise is Gumbel-distributed, the probability of selecting a specific discrete channel θ reduces to Luce's choice ratio:

$$P_\theta = \frac{\exp\Big(U_\theta\Big)}{\sum_{\phi=1}^{360} \exp\Big(U_\phi\Big)} \tag{1}$$

where P_θ represents the discrete probability mass of responding to a given detector channel. To express these selection probabilities as a continuous probability density distribution over a circular space, we normalize the choice rule by the detector bin width, $h = \frac{2\pi}{360}$. The continuous density function L for any item-centered error ϵ is:

$$L(\epsilon; A, \kappa) = \frac{\exp\Big(U(\epsilon)\Big)}{\sum_{\phi=1}^{360} \exp\Big(U(\theta_\phi)\Big) \cdot h} \tag{2}$$

Neural tuning and activation, $U(\epsilon)$, are given by a scaled von Mises function centered on the item's true feature value:

$$U(\epsilon) = \frac{A}{2\pi} \exp\Big(\kappa \cos(\epsilon)\Big) \tag{3}$$

where A represents the peak amplitude and κ represents detector concentration (tuning sharpness). Additionally, we allowed for swap errors using a two component mixture model, where the final probability density for response error e at size N is given by:

$$p(e \mid A, \kappa, \gamma) = (1-\gamma) L(\epsilon_{\text{target}}; A, \kappa) + \gamma \frac{1}{N-1} \sum_{j=1}^{N-1} L(\epsilon_j; A, \kappa) \tag{4}$$

where γ is the swap rate, ϵ_{target} is the response error relative to the target, ϵ_j is the response error re-centered relative to the j-th non-target distractor, and N is either the number of items (a location-based swap error) or the number of colors (a memory based swap error).

Hypothesis testing

We used a nested forward-selection strategy to test our main hypotheses: Gestalt encoding, probability summation, and neural summation/selective attention (Table 2). For each participant, the SDM provided a baseline 'best fit' account with $9A$ (one per sub-condition) and a shared κ. We then assessed Gestalt encoding by allowing three amplitude parameters to vary by the number of array items (N_i: 2, 4 or 6) or array colors (N_c: 2, 4 or 6). If the unit of visual working memory is defined by the number of items in a memory array, then recall amplitude will progressively decline with N_i. If however, redundant colors form a unitary Gestalt, then the number of unique colors, N_c will define the unit of VWM and describe how amplitude declines across our nine sub-conditions. Model comparison will determine which account best fits the data and therefore define N in our subsequent models.

Table 2
Experiment 2 hypotheses and model class tests. The SDM (Oberauer, 2023) provides a baseline 'best fit' with one A per sub-condition. Gestalt encoding (H_1): allowed A to vary with the number of items (N_i) or number of colors (N_c) and subsequently defined N. Probability summation (H_2): allowed recall precision to decline with one β either with or without redundant item probability summation (where m is the number of non-redundant colors in the array). Conditional attention weights (H_3): freed or fixed β to sample-size in Redundant, Non-Redundant and Unique item conditions and assessed neural summation, attentional trade-off and spill-over effects; probability summation was applied if supported by H_2. A theoretical A_1 was estimated from $A_2 2^{-\beta}$ in all β models. Swap models were assessed during H_3 model comparison.

	Model Class	Free parameters	Rule
	A Baseline (SDM)	$A_{1...9}$, κ	Free A, fixed κ
H_1	*Gestalt Encoding Hypothesis*		
	Gestalt Encoding (N_c)	A_1, A_2, A_3, κ	A on N_c, fixed κ
	Item Encoding (N_i)	A_1, A_2, A_3, κ	A on N_i, fixed κ
H_2	*Redundant Item Probability Summation*		
	No Summation	A, κ, β_{all}	$A_N = A2^{-\beta}N^{\beta}$
	Redundancy Summation	A, κ, β_{all}	$A_N = A2^{-\beta}m^{-\beta}N^{\beta}$
H_{2a}	*Attentional Weights (R, NR, Unique)*		
	Power-Law on A	A, κ, β	H_{2a}
	Sample-Size on A	A, κ, $\beta = -0.5$	$A_N = A\sqrt{2/N}$
	Sample-Size on A w/ Prob. Sum.	A, κ, $\beta = -0.5$	$A_N = A\sqrt{2m/N}$

Next we assessed whether probability summation should be used as an assumed

baseline model of redundant item activation. To simplify the presentation of this test, we compared fits for a 1β model wherein redundant item recall was or was not boosted via probability summation. If redundant item amplitudes (i.e., signal-to-noise) are boosted due to probability summation, then their closest baseline model will follow an $(m/N)^{\beta}$ relationship and non-redundant and unique item sets will follow a $1/N^{\beta}$ relationship. However, if redundant items do not experience a sufficient amplitude boost, then all conditions will be best fit by a $1/N^{\beta}$ relationship. We subsequently confirmed that the conclusions of this simplified model test held when β was allowed to vary by redundant, non-redundant and unique item conditions; all model tests are reported in Appendix B. If a participant was best fit assuming probability summation, then we used this as the baseline prediction from which to infer attention reallocation and neural summation.

Informed as to the underlying unit of VWM, N, and whether to assume redundant item probability summation as our baseline prediction, we next assessed how conditional attention differed relative to our baseline model predictions. We fit independent β models to redundant, non-redundant and unique item conditions, either while freeing conditional βs (power-laws) or fixing them to sample-size model predictions (β = -0.5). This resulted in six model variants (see Appendix B). The most saturated model variant was a free 3β $[\beta_R, \beta_{NR}, \beta_U]$, $1A, 1\kappa$ model. The most constrained variant was a $1A, 1\kappa$ sample-size ($\beta_{all} = -.5$) model.

Assuming probability summation, then a redundant item β_R that out-performs sample-size (i.e., power-law) predictions implies either neural summation or attention reallocation. If non-redundant items (β_{NR}) match the sample-size model, or at minimum the predictions of unique color sets (β_U, assuming unique items perform worse than sample-size), then redundant items interact minimally with non-redundant elements in a mixed array. This outcome aligns with neural summation. However, if β_{NR} performs worse than β_U, the model suggests an attentional trade-off: redundant item performance is benefiting by siphoning attention from non-redundant elements.

If our model tests do not support probability summation, then the subsequent β

model cannot be used to infer neural summation. When modeled against an independent activation baseline, redundant item performance is likely to reflect a mixture of two processes: independent item activations and activations arising from a small proportion of probability summation. Although this probability summation was insufficient to win during model comparison, *any* such occurrence will result in an amplitude activation that exceeds an independent sample-size baseline model (as one would expect per neural summation). Fortunately, in these circumstances, meaningful insights may still be gleamed from the non-redundant and unique item attention weights.

Models were fit using maximum-likelihood estimation via the *fmincon* function in Matlab 2023b with 100 random starts sampled uniformly from within lower- and upper-bounds. Amplitude was free to vary from 0-30; κ from 0-15, β from -1 to 0 (restricted from ± 2 after initial testing) and swap probabilities from 0-1. We use BIC for model comparison as it penalizes model complexity and is better suited when selecting among theoretical accounts. We report AIC as a prediction-oriented fit criterion. We simulate data from the best constrained β model fits to compare circular standard error predictions to empirical data, and show model fits as distributional probability density functions relative to both the empirical data and the free SDM model.

Results and Discussion

Experiment 1

We excluded six participants who completed only one experimental session and seven more after mixture-model analysis (Appendix Table A1). Across participants, three trials were excluded for recording error, 11 for being too fast, and 475 for being too slow. The final dataset comprised 21 participants (M_{trials} = 793, SD = 38).

RT. In keeping with the Systems Factorial Technology literature, we expected RT redundancy gains to scale as the number of redundant elements increased (e.g., Thornton & Gilden, 2007). Empirical data supported a single-color redundancy gain in

RT relative to other arrays, no redundancy gain within mixed arrays.

Figure 4 (top) displays Experiment 1 mean RTs with 95% HDI error-bars. Conditions on the x-axis show unique (1, 2, 4, 6) and mixed redundant (R) and non-redundant arrays (NR), with subscript numbers and marker color denoting color-count. Mean RT increased monotonically with unique colors ($BF_{10} = 1.1 \times 10^{11}$; $F(1.43, 28.7) = 39.1$, $p < .001$). Post-hoc comparisons showed RT for single-color arrays were faster than all other sets (BF_{10}s $> 7.5 \times 10^3$; $p < .001$); however, displaying more single-colored items did not improve redundant item response speed ($BF_{10} = 0.29$; $F(2.85, 57.09) = 1.45$, $p = .24$).

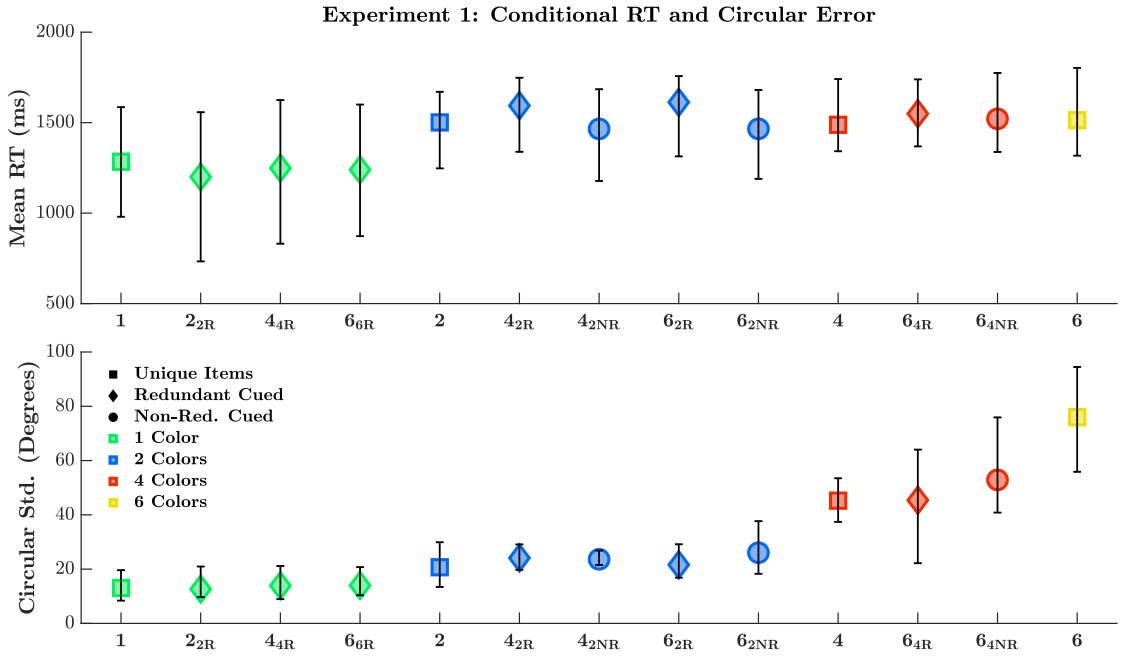


Figure 4. Mean RT (top) and mean circular standard deviation (bottom) across participants for Experiment 1 sub-conditions ($ItemN_{ColorN}$). Error bars display non-parametric narrowest 95% highest density intervals, calculated through a double-bootstrap procedure; differences may be inferred where error bars do not cross.

Multi-color array RTs did not support redundancy gains. Two-color non-redundant RTs were faster than redundant and unique displays within four- and six-item sets (set-size + cue-type model: $BF_{10} = 5.08 \times 10^6$; $F(1, 20) = 10.3$, $p < .01$; main effect set-size: $F(1, 20) = 18.55, p < .001$, main effect cue-type: $F(2, 40) = 5.9, p < .01$; set-size $\times$ cue-type interaction: $F(2, 40) = 6.42, p < .01$). Post-hoc tests were only significant for non-redundant comparisons (all $BF_{10} > 2700$; p

$< .05$). This effect diminished at six items, showing no difference between unique and four-color (non)redundant cue conditions ($BF_{10} = 0.38$; $F(1, 40) = 1.49$, $p = .24$). Conditional HDIs suggest large individual differences for all RT effects.

Precision. At the group level, we expect recall precision – the inverse of circular standard deviation – to display a redundancy gain. Figure 4 (bottom) displays Experiment 1 mean circular SD with 95% HDIs. Circular standard deviation increased (i.e., precision decreased) across unique item-sets ($BF_{10} = 8.17 \times 10^3 1$; $F(1.91, 38.26) = 210$, $p < .001$); all post-hoc comparisons were significant (all $BF_{10} > 14 \times 10^3$; $p < .001$). Precision did not increase with the number of items shown in single-color arrays ($BF_{10} = 0.106$; $F(12.2, 544.3) = 0.45$; $p = .7$).

Redundant and non-redundant two-color precision improved relative to unique arrays (set-size + cue-type model: $BF_{10} = 1.35 \times 10^{31}$; main effect item-set: $F(1, 20) = 60.9$; $p < .001$, main effect cue-type: $F(1, 20) = 152.8$; $p < .001$, set-size × cue-type interaction: $F(2, 40) = 75.6$; $p < .001$). This was supported by post-hoc comparisons (BF_{10}s $> 7.49 \times 10^{12}$; $p < .001$). No difference was found between redundant and non-redundant sets ($BF_{10} = .168$; $p = 1$). An identical pattern was found among the four-color conditions ($BF_{10} = 9.08 \times 10^9$; $F(2, 40) = 57.9$, $p < .001$); all post-hoc comparisons were significant (all $BF_{10} > 231$; $p < .001$). Results suggest that the number of colors determines memory recall capacity, and that redundancy may benefit the recall of both redundant and non-redundant elements when compared to recall performance for the same number of unique colors.

Interim Discussion

When shown more items within the single-color condition, additional redundancy did not produce an RT or precision benefit. However, single-color arrays were faster and more precise than mixed arrays. We suspect that, when presented with a single color, participants committed to their response during stimulus presentation, thereby producing faster and more precise single-color responses.

In multi-color arrays, redundant item precision improved but RT slowed, implying

a performance trade-off. Non-redundant precision and RT also improved. This may suggest a 'spill over' benefit for non-redundant items; however, this may be an averaging artifact. Large HDIs suggest individual differences in recall strategies. It is possible some strategies improve redundant item recall, while others improve non-redundant item performance: On average, this would appear as a 'spill over' benefit.

The importance of these group-level results is three fold. Firstly, to identify manipulations that are not conducive to further study – single-color set-size manipulations. Secondly, to show where robust psychological phenomena persist – color rather than item based capacity limits in visual working memory precision. Thirdly, to show how theoretical mistakes can be made when averaging over individual differences. To interrogate our third point, we next examine individual precision model fits in Experiment 2.

Experiment 2

Across five participants, six trials were excluded due to recording errors, one trial for being too fast, and 23 trials for being too slow. All participants passed mixture model screening (appendix Table A2). On average, 4492 (SD = 4) trials were analyzed for each participant.

Mean RT. Figure 5 displays participants' conditional mean RTs with non-parametric 95% HDI error-bars. In brief, RTs increased monotonically with unique item arrays, however, as implied by group HDIs, redundancy effects differ by individual. Of note, P03 – P05 showed a non-redundant RT benefit when compared to redundant cues, which was absent in P01 – P02.

Precision Modeling. To avoid redundancy (pun intended) in the presentation of our results, we present model fits and precision means together. Table 3 shows results for the Gestalt and probability summation model tests. Gestalt hypothesis tests supported color as the underlying unit of visual working memory. This means our subsequent models defined N by the number of colors within a memory array. Probability summation was supported in three of five participants (P01, P02, P04).

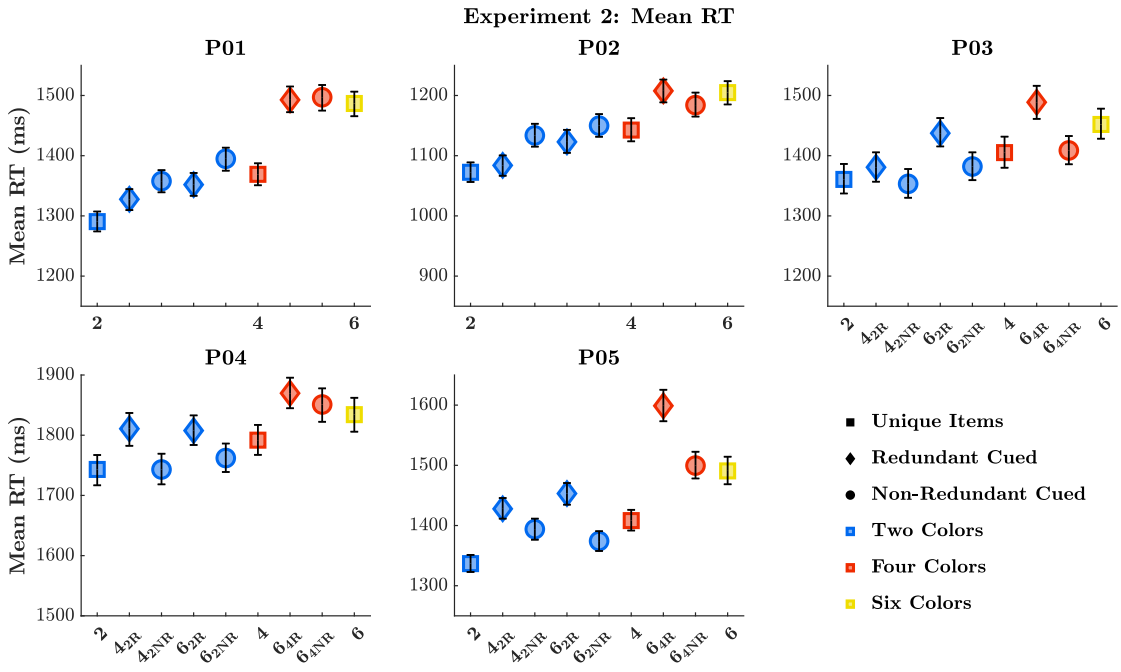


Figure 5. Mean RT per condition (x-axis) and person (subplots) in Experiment 2. Error bars display non-parametric narrowest 95% highest density intervals; differences may be inferred where error bars do not cross.

This means β models for these participants were fit with probability summation as their baseline prediction for redundant item performance. Performance exceeding this probability summation baseline may indicate an attentional bias or neural summation account of redundancy. By contrast, redundant item precision for participants P03 and P05 did not support probability summation: their redundant item baseline remained a $1/N^{\beta}$ relationship. Performance beyond this independent activation baseline does not support neural summation, as it likely reflects a mixture of probability summation and independent neural activations across trials, and therefore predicts better-than independent-activation sample-size performance.

Table 4 displays the best β models for each participant. Model order progresses top-to-bottom from the free A baseline (i.e., the SDM), the best fitting β model without swap errors, then with swap errors conditioned on the number of colors (Swap_c), and then swap errors on the number of items (Swap_i). Throughout model fitting, swap component probabilities dropped to zero when conditioned on two and four items, or two and four colors. As such, we only report swap errors when conditioned on six items

Table 3
Model comparison tests for the Gestalt and Probability Summation hypotheses. Model comparison supports Gestalt encoding in all participants, implying that the number of colors within an array defines the unit N of visual working memory. Model comparison also supports probability summation accounts in three of five participants.

ID	Model	LL	AIC	BIC
H_1 **Gestalt Hypothesis Tests**				
P01	Free A	2834	5687	5752
P01	A on Color N	3006	6021	**6046**
P01	A on Item N	3206	6420	6446
P02	Free A	2270	4559	4624
P02	A on Color N	2323	4655	**4680**
P02	A on Item N	2364	4736	4762
P03	Free A	2453	4925	4990
P03	A on Color N	2496	5001	**5026**
P03	A on Item N	2778	5564	5590
P04	Free A	2118	4256	4321
P04	A on Color N	2172	4353	**4379**
P04	Item N	2340	4688	4713
P05	Free A	1939	3897	3961
P05	A on Color N	1954	3915	**3941**
P05	A on Item N	2141	4290	4315
H_2 **Probability Summation Tests**				
P01	Probability Summation	2896	5798	**5818**
P01	No Probability Summation	3014	6034	6053
P02	Probability Summation	2286	4577	**4596**
P02	No Probability Summation	2326	4659	4678
P03	Probability Summation	2567	5139	5158
P03	No Probability Summation	2506	5018	**5037**
P04	Probability Summation	2151	4309	**4328**
P04	No Probability Summation	2181	4368	4387
P05	Probability Summation	2095	4196	4216
P05	No Probability Summation	1958	3923	**3942**

or six colors – a point for later discussion. All constrained model fits for P02 - P05 outperformed the baseline SDM on BIC. P01, P03 and P05 favored a swap model component conditioned on the number of item set colors.

Table 4 interpretations are clear but nuanced. P01 and P03 display a redundancy trade-off: β_R performs better than sample-size baseline predictions ($\beta = -.5$) while β_{NR} displays a severe capacity deficit. In both cases, β_{NR} lower than β_U. This means non-redundant attention is being selectively affected when presented alongside redundant items: an attentional trade-off. We note that P01 supported probability summation, while P03 did not, and that unique items were fit by a strict sample-size rule for P01 (β_U = -0.5) whereas P03 was worse-than sample-size (β_U = -0.67). Both

Table 4
Free beta and swap model fits. Beta models for P01, P02, and P04 were fit with probability summation ($\beta_R = \sqrt{m}/\sqrt{N}$); P03 and P05 were fit without probability summation ($\beta_R = 1/\sqrt{N}$). Model interpretations: P01 and P03 display a redundancy trade-off; P02 displays a universal set-size benefit; P04 displays neural summation; P05 displays a non-redundancy trade-off (attending non-redundant elements at a cost to redundant item performance). Swap probabilities conditioned on six-item color count ($Swap_c$) benefited P01, P03 and P05 BIC. Bold BIC denotes best fitting models; Green rows denote best fitting β models.

ID	**Model**	**LL**	**AIC**	**BIC**	$\mathbf{P}_n$	**A**	**K**	$-\beta_R$	$-\beta_{NR}$	$-\beta_U$	$\mathbf{Swap}_p$
P01	Free A	2834	5687	**5752**	10	[5.2 - 1.7]	2.13				
P01	3β: R, NR, U	2878	5765	5798	5	3.43	2.28	0.41	0.92	0.62	
P01	2β: R, NR, U=.5 ($Swap_c$)	2868	5746	5778	5	3.79	2.19	0.42	0.93	0.5	0.18
P01	3β: R, NR, U ($Swap_i$)	2874	5759	5797	5	4.05	2.15	0.42	0.94	0.64	0.02
P02	Free A	2270	4559	4624	10	[4.6 - 2.7]	2.22				
P02	1β: R = NR = U	2286	4577	**4596**	5	3.57	2.26	0.24	0.24	0.24	
P02	1β: R = NR = U ($Swap_c$)	2281	4571	4597	6	3.9	2.2	0.22	0.22	0.22	0.06
P02	1β: R = NR = U ($Swap_i$)	2284	4575	4601	6	3.9	2.2	0.24	0.24	0.24	0.01
P03	Free A	2453	4925	4990	10	[0.6 - 0.2]	4.03				
P03	2β: R, NR = U	2471	4949	4975	4	0.64	4.08	0.39	0.87	0.87	
P03	3β: R, NR, U ($Swap_c$)	2457	4926	**4965**	6	0.65	4.04	0.38	0.98	0.67	0.24
P03	2β: R, NR = U ($Swap_i$)	2468	4946	4978	5	0.7	3.99	0.33	0.87	0.87	0.02
P04	Free A	2118	4256	4321	10	[2.3 - 1.8]	2.76				
P04	1β: R, [NR,U] = .5	2127	4260	**4279**	3	2	2.88	0.25	0.5	0.5	
P04	1β: R, [NR,U] = .5 ($Swap_c$)	2123	4255	4280	4	2.26	2.78	0.2	0.5	0.5	0.08
P04	1β: R, [NR,U] = .5 ($Swap_i$)	2123	4254	4279	4	2.44	2.7	0.23	0.5	0.5	0.01
P05	Free A	1939	3897	3961	10	[1.37 - 0.57]	3.46				
P05	1β: R = NR = U	1958	3923	3942	3	1.26	3.49	0.69	0.69	0.69	
P05	1β: R, [NR,U] = .5 ($Swap_c$)	1939	3886	**3912**	4	1.4	3.4	0.81	0.5	0.5	0.19
P05	1β: R = NR = U ($Swap_i$)	1958	3924	3950	4	1.29	3.47	0.69	0.69	0.69	<.01

supported six-color swap errors.

Table 4 results for P04 but not P02 support neural summation. In both participants, β_R performed better than a probability summation sample-size baseline, with β_{NR} and β_U performing at or better-than sample-size (i.e., no redundancy trade-off). The findings for P04 align with neural summation (better-than sample-size β_R with no attention trade off); however, P02 is better explained by an alternative theory. In their post-experiment interview, P02 described actively "*grouping arrays by color and shape*", a strategy that could produce better-than sample-size performance across redundant, non-redundant and unique array conditions. For example, colored-shapes may form configural ensembles, further lowering the unit N of VWM and violating our baseline sample-size model predictions. Rather than a redundancy 'spill-over' benefit – which would benefit non-redundant items but not unique arrays – configural strategies would compresses array elements into structural Gestalts regardless

of their redundant or unique color status. In turn, this would predict better-than sample-size model performance in all conditions.

Finally, Table 4 results for P05 display a unique instance of non-redundant item trade-off – where redundant items perform worse than their non-redundant counterparts. In their post-experiment interview, P05 described "*intentionally attending to non-redundant colors ... to counteract redundancy*". This strategy was captured in the model, with β_{NR} and β_U perfectly matching sample-size predictions, and β_R displaying *worse than* sample-size performance (i.e., a severe capacity limitation). This strategic decision may explain why P05 did not display redundant item probability summation.

Figure 6 displays the best constrained β Model fits to empirical circular standard error in each sub-condition. Per our Gestalt hypothesis, precision is clearly responsive to the number of colors within each display, rather than the number of items. The subtle influence of probability summation can also be noted by comparing the relative placement of redundant-conditions (diamonds) in P01, P02 and P04, to the non-summation accounts of P03 and P05. Redundant item costs are also clearly illustrated for P05 in the four-color conditions (red markers).

Figure 7 displays P01's best distributional fits for the Free A model (the SDM) and constrained β model, overlaid on empirical data. Of the five participants, P01 was worst fit by a β model account with distributional fits missing peaks at set sizes 2 and 6. Even so, their model fits substantively capture the data across nine sub-conditions to provide a clear interpretation of participant's attentional strategies.

Figure 8 displays P05's best distributional fits for the Free A and constrained β model, overlaid on empirical data. Of our five participants, P05 was best fit by their β model, with comparable log-likelihood and AIC performance to the Free A model across all nine sub-conditions. The remaining distributional fits are qualitatively and quantitatively similar to P05. For exposition, we provide these as Appendix Figures C1, C2, and C3.

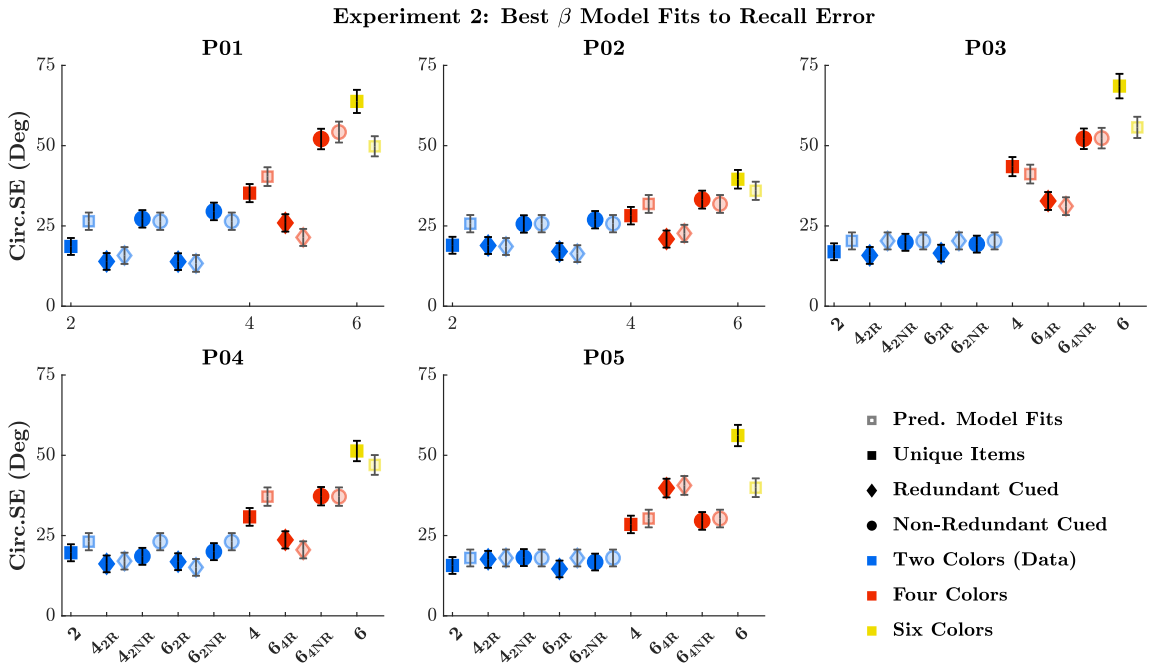


Figure 6. Best fitting Experiment 2 β models plotted against empirical data for each participant (subplot) and condition (x-axis). Transparent markers are model fits. Colors denote the number of unique colors; diamond, circle, and squares denote redundant, non-redundant and unique item conditions.

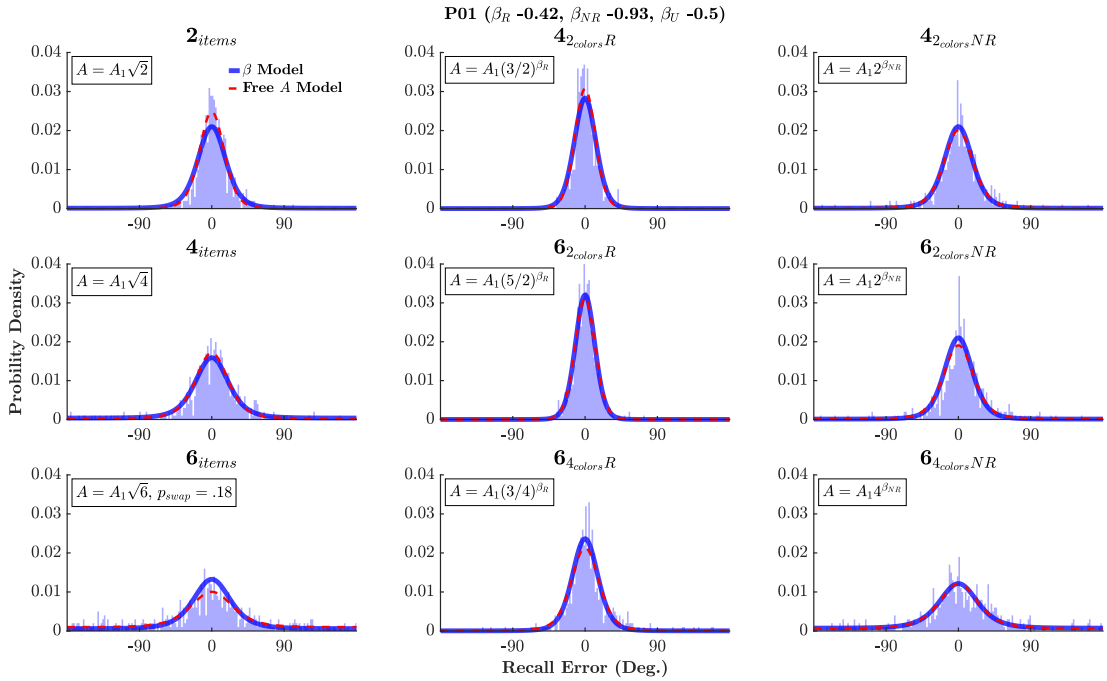


Figure 7. P01 empirical histograms overlaid by a 2β probability summation + swap model (blue-line) and free amplitude model fits (red-dash). Annotations denote conditional amplitude calculations. N is color count, redundant conditions (β_R) benefit from probability summation, non-redundant items follow a separate power-law (β_{NR}), and unique conditions follow a sample-size square-root law.

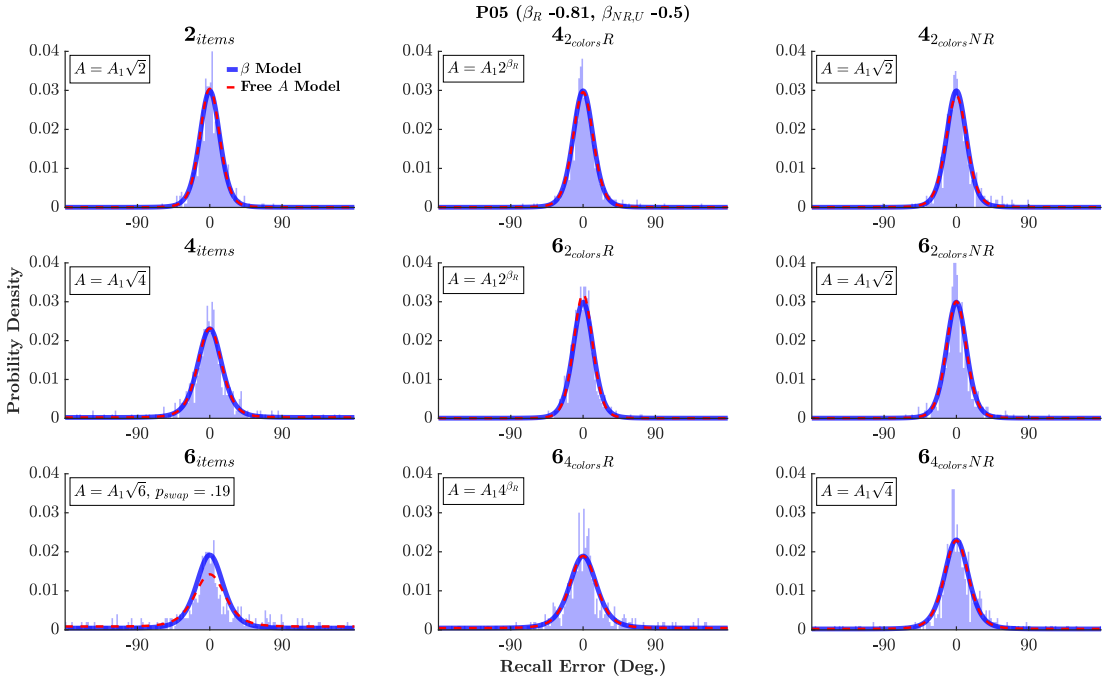


Figure 8. P05 empirical histograms overlaid by a 1β swap model without probability summation (blue-line) and free amplitude model fits (red-dash). Annotations denote conditional amplitude calculations. N is color count, redundant conditions follow a β_R power-law, while unique and non-redundant conditions follow a sample-size rule.

General Discussion

We set out to assess the consequences of redundancy for visual working memory using the sample-size model of neural activation, and the tightly constrained predictions of Gestalt encoding, probability summation, and neural summation/attention reallocation. In Experiment 1, participants recalled a single item from arrays containing either uniquely colored items or a mixture of redundant and non-redundant items. Group results supported Gestalt encoding – the number of unique colors defined the unit of VWM – but individual differences masked further theoretical interpretations. In Experiment 2, individual model fits universally supported Gestalt encoding and in three participants, probability summation. Model fits further highlighted individual VWM strategies: Attentional reallocation both towards (P01, P03) and away from (P05) redundant items, redundant item neural summation (P04) and universal color-shape Gestalt encoding (P02). We posit that group-level redundancy results in Experiment 1 reflect a similar mixture of individualized strategies. Our results highlight the

importance of tightly constrained, individualized modeling when theory crafting VWM experiments, especially when studying the effects of redundancy.

Redundancy gains in recall precision

Gestalt encoding emerged as the underlying unit of VWM in all participant model fits (Experiment 2), and was robustly observed in group-level analyses (Experiment 1). This finding contributes to a growing body of literature showing VWM capacity is defined by ensemble 'objects' (i.e., unitary color regularities) rather than individuated items. This higher-order ensemble encoding improves memory precision for both redundant and non-redundant elements within the same array.

Probability summation defined redundant item baseline predictions in three of five participants. Of these, one participant supported neural summation, one attention reallocation away from non-redundant elements, and one a configural Gestalt advantage regardless of item condition. P03 did not display probability summation but did show a redundancy benefit due to attention reallocation. We suspect this redundancy gain also reflects a stochastic mixture of independent and summed neural firing. P05 did not display probability summation, likely due to their decision to focus on non-redundant colors. Our observed differences in probability summation across participants capture the non-obligatory, post-encoding (second-order) nature of this phenomenon.

Our findings resolve a theoretical tension between capacity-limited and array ensemble phenomena in VWM. Our combined neural-population and sample-size framework reconciles divisive normalization and attention weightings with 'limited capacity' with 'ensemble sensitivity'. Redundant features boost global capacity limits by allowing memory to store color-bound unitary objects. Simultaneously, redundant features can elevate sensitivity via probability summation, with deviations from this baseline indexing either attention reallocation, neural summation, or violations of perceptual independence.

As mentioned in the Introduction, our baseline model predictions necessarily assumed items are not interacting (e.g., forming shapes or pattern Gestalts). This is a

necessary simplifying assumption in order to project how single-item baseline amplitude should theoretically scale across set sizes. While predicted violations of this simplifying assumption, as occurs with neural summation, become theoretically diagnostic, unpredicted violations, as in P02, require speculation. Model fits and self reports suggest P02 engaged in configural encoding and retrieval for redundant, non-redundant and uniquely colored item sets. For example, P02 described associating colors with shapes and patterns to enhance memory retrieval above our baseline model predictions. We suspect this strategy requires highly efficient attentional scanning and intentional deployment, limiting its use among our participants. Future research may use eye tracking while instructing participants towards these theoretically efficient recall techniques to determine if such beneficial expertise can be taught and applied.

Swap errors. Model fits favored the inclusion of swap errors in three of five participants. We expected swap errors to boost redundant item recall without penalizing non-redundant elements – a location-bound swap effect. However, model results strictly favored color-bound swap error and only in the six-color condition. Color-bound swap errors may reflect confusions in memory: Gestalt encoding identifies color as the VWM unit, so memory swaps would occur between unique colors not between item locations. However, another interpretation posits that six-color swap errors merely reflect model flexibility.

Of the nine sub-conditions we assessed, only one included a six-color item-set. This makes the benefits of the swap parameter both highly targeted and unrestricted by other conditions. Fortunately, participant β models do not meaningfully change when swap errors are excluded. The only notable effect occurs in P05 whose model shifts to a single-β = -.69 account. Although entirely plausible, this result is notably less in keeping with their self-reported strategy. For now, we remain open to the notion that swap errors may occur at the level of the VWM unit – color in this study – and consider it an interesting avenue of future study.

Future sample-size insights. The sample-size model of neural activation affords potential insights into capacity bound phenomena. In standard VWM tasks,

recall precision should follow standard sample-size predictions. Violations of this baseline model therefore point to unexplained phenomena: Swap errors, summation gains, ensemble or object-based encoding, and attentional biases. In theory, a model that therefore accounts for these structures should observe a shift back towards the predicted sample-size baseline, making the baseline model predictions incredibly useful when theory testing. In more complex attentional tasks, as in the current redundancy design, violations away from sample-size can instead become useful hypothesis tests – as we showed in this paper. In keeping with the theme of this paper, we contend that the value of this model is not to be 'the best fitting' but instead, to be extremely principled in its predictions and in how researchers can use these predictions to create and test their theories. In principle, sample-size predictions should extend across population coding frameworks, allowing the same baseline logic to be incorporated under different model assumptions. For example, psychological scaling under the target confusability competition model (Schurgin et al., 2020), or discretized versus continuous resource allocation under the stochastic sampling model (Schneegans et al., 2020).

Future response time insights. While this paper has focused on precision effects, we note clear RT differences that would be suitable for joint modeling. Recently, Smith et al. (2025) jointly modeled recall RT and precision in a color-recall task using the *Circular Diffusion Model*; although the *Spatially Continuous Diffusion Model* (Fennell & Ratcliff, 2023) provides an equally viable alternative (see Smith et al., n.d, for direct comparisons). Both frameworks can incorporate our sample-size model by mapping population amplitude to drift rate and tuning width to drift variability. We contend that requiring neural-population models to predict RT *and* precision together may offer a principled way to distinguish theoretical accounts that appear similar when judged on precision alone (e.g., Garrett et al., 2026).

Conclusion

The attention-weighted sample-size model of neural activation substantially explains recall precision across unique, redundant and non-redundant color arrays. Our

highly constrained and theoretically insightful combination of the SDM population coding framework and sample-size model resolves a theoretical quandary posed by minimally interacting limited capacity models and ensemble array statistics by treating color as the unit of VWM capacity. Our theoretically principled baseline predictions provide a baseline prediction for probability summation, and a theoretical floor on which to establish neural summation, attentional trade-offs, and theoretical benefits (e.g., spill-over and configural effects). We conclude that VWM redundancy gains confer nuance and individualized effects that show enormous potential for scientific manipulation and theoretical development in the study of attention, memory and decision making.

Appendix A

Mixture model analyses

Table A1

Uniform and von Mises mixture model results for Experiment 1. Bold IDs were removed from further analyses due to exceeding 10% or 20% uniform responding in set sizes 1 and 2, respectively. Proportions are the uniform mixture components.

ID	N_{1p}	N_{2p}	N_{4p}	N_{6p}	Kappa	ID	N_{1p}	N_{2p}	N_{4p}	N_{6p}	Kappa
E01	0.75	0.85	0.84	0.8	4.04	E15	0	0.01	0.03	0.13	22.77
E02	0	0	0.17	0.26	16.41	E16	0	0	0.05	0.13	14.13
E03	0.12	0.07	0.26	0.34	11.39	E17	0	0.01	0.11	0.19	6.68
E04	0	0	0.13	0.2	11.83	E18	0	0.01	0.14	0.23	22.18
E05	0	0.04	0.13	0.3	10.99	**E19**	0	0.32	0.42	0.52	12.02
E06	0	0.03	0.12	0.26	16.48	E20	0.04	0.13	0.31	0.42	12.82
E07	0.13	0.42	0.59	0.63	15.63	E21	0	0.05	0.2	0.29	7.54
E08	0	0	0.12	0.23	11.64	E22	0	0.02	0.17	0.31	10.8
E09	0	0.05	0.12	0.34	7.52	**E23**	0.17	0.33	0.48	0.5	7.75
E10	0.13	0.16	0.44	0.53	9.89	E24	0	0	0.07	0.14	9.66
E11	0	0.11	0.24	0.37	11.87	E25	0	0.04	0.06	0.14	16.04
E12	0	0.05	0.11	0.16	10.54	E26	0	0.07	0.12	0.29	13.9
E13	0.05	0.27	0.28	0.49	5.58	E27	0	0.02	0.11	0.23	13.48
E14	0.08	0.05	0.16	0.3	10.26	E28	0	0.03	0.11	0.25	11.33

Table A2

Uniform and von Mises mixture model results for Experiment 2. Proportions are the uniform mixture components.

ID	N_{2p}	N_{4p}	N_{6p}	Kappa
P01	0.01	0.07	0.2	9.44
P02	0	0.04	0.08	9.26
P03	0.03	0.11	0.23	15.77
P04	0.01	0.05	0.13	11.52
P05	0.01	0.06	0.16	14.6

Appendix B

Population coding model fits

Table B1

Beta model fits without a six-color swap error component, where redundant item probability summation either was (left), or was not included (right). All models assume color count, N, to be the underling unit of VWM. Raw model outputs are provided on the OSF (osf.io/pxskw).

		Probability Summation			**No Probability Summation**		
ID	**Model**	**LL**	**AIC**	**BIC**	**LL**	**AIC**	**BIC**
P01	0β: Sample Size $\beta_{all} = .5$	2897	5799	5812	3041	6087	6100
P01	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2897	5799	5819	3012	6030	6050
P01	1β: Power-law $1\beta_{all}$	2896	5798	5818	3014	6034	6053
P01	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2886	5780	5806	2964	5936	5962
P01	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2882	5772	5798	2979	5965	5991
P01	3β: β_R, β_{NR}, β_U	2878	5765	5798	2954	5917	5949
P02	0β: Sample Size $\beta_{all} = .5$	2365	4735	4747	2338	4680	4693
P02	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2297	4599	4618	2310	4626	4645
P02	1β: Power-law $1\beta_{all}$	2286	4577	4596	2326	4659	4678
P02	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2284	4576	4601	2306	4619	4645
P02	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2297	4601	4627	2310	4627	4653
P02	3β: β_R, β_{NR}, β_U	2283	4576	4608	2304	4618	4650
P03	0β: Sample Size $\beta_{all} = .5$	2574	5152	5165	2542	5088	5101
P03	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2502	5010	5030	2527	5060	5080
P03	1β: Power-law $1\beta_{all}$	2550	5105	5124	2503	5012	5032
P03	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2482	4971	4997	2471	4949	4975
P03	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2496	5000	5026	2512	5031	5057
P03	3β: β_R, β_{NR}, β_U	2481	4971	5003	2469	4947	4979
P04	0β: Sample Size $\beta_{all} = .5$	2178	4360	4373	2187	4378	4391
P04	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2127	4260	4279	2156	4317	4336
P04	1β: Power-law $1\beta_{all}$	2151	4309	4328	2181	4368	4387
P04	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2124	4256	4282	2141	4291	4316
P04	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2126	4260	4286	2152	4312	4337
P04	3β: β_R, β_{NR}, β_U	2124	4258	4290	2141	4292	4324
P05	0β: Sample Size $\beta_{all} = .5$	2208	4420	4433	1974	3951	3964
P05	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2058	4122	4142	1965	3936	3956
P05	1β: Power-law $1\beta_{all}$	2089	4185	4204	1958	3923	3942
P05	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2057	4123	4148	1954	3916	3942
P05	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2051	4110	4136	1964	3937	3963
P05	3β: β_R, β_{NR}, β_U	2051	4112	4144	1951	3911	3943

Table B2

Beta model fits that include a six-color swap error component. Redundant item probability summation either was (left), or was not included (right). All models assume color count, N, to be the underling unit of VWM. Raw model outputs are provided on the OSF (osf.io/pxskw).

		Probability Summation			**No Probability Summation**		
ID	**Model**	**LL**	**AIC**	**BIC**	**LL**	**AIC**	**BIC**
P01	0β: Sample Size $\beta_{all} = .5$	2889	5784	5803	3019	6044	6063
P01	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2888	5784	5810	2995	5998	6024
P01	1β: Power-law $1\beta_{all}$	2889	5786	5812	3008	6025	6050
P01	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2885	5779	5811	2964	5938	5970
P01	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2868	5746	5778	2952	5913	5945
P01	3β: β_R, β_{NR}, β_U	2868	5747	5786	2948	5908	5947
P02	0β: Sample Size $\beta_{all} = .5$	2365	4737	4756	2338	4682	4701
P02	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2297	4601	4627	2310	4628	4653
P02	1β: Power-law $1\beta_{all}$	2281	4571	4597	2321	4649	4675
P02	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2281	4573	4605	2305	4619	4651
P02	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2297	4603	4635	2310	4629	4661
P02	3β: β_R, β_{NR}, β_U	2278	4568	4606	2301	4613	4652
P03	0β: Sample Size $\beta_{all} = .5$	2557	5121	5140	2498	5003	5022
P03	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2477	4962	4988	2489	4986	5012
P03	1β: Power-law $1\beta_{all}$	2507	5022	5047	2488	4983	5009
P03	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2473	4956	4989	2466	4941	4973
P03	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2466	4941	4973	2463	4936	4968
P03	3β: β_R, β_{NR}, β_U	2465	4943	4981	2457	4926	4965
P04	0β: Sample Size $\beta_{all} = .5$	2176	4358	4377	2175	4357	4376
P04	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2123	4255	4280	2149	4305	4331
P04	1β: Power-law $1\beta_{all}$	2139	4286	4312	2175	4357	4383
P04	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2123	4256	4288	2141	4292	4324
P04	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2122	4253	4285	2142	4295	4327
P04	3β: β_R, β_{NR}, β_U	2122	4255	4294	2140	4291	4329
P05	0β: Sample Size $\beta_{all} = .5$	2205	4415	4435	1954	3914	3933
P05	1β: $1\beta_R$, $[\beta_{NR}, \beta_U] = .5$	2048	4104	4130	1939	3886	3912
P05	1β: Power-law $1\beta_{all}$	2044	4096	4122	1950	3908	3934
P05	2β: $1\beta_R$, $1\beta_{NR} = \beta_U$	2037	4083	4115	1939	3888	3920
P05	2β: $1\beta_R$, $1\beta_{NR}$, $\beta_U = .5$	2044	4097	4129	1939	3888	3920
P05	3β: β_R, β_{NR}, β_U	2036	4084	4123	1939	3890	3928

Appendix C

Exp 2 Amplitude Model Distributional Fits

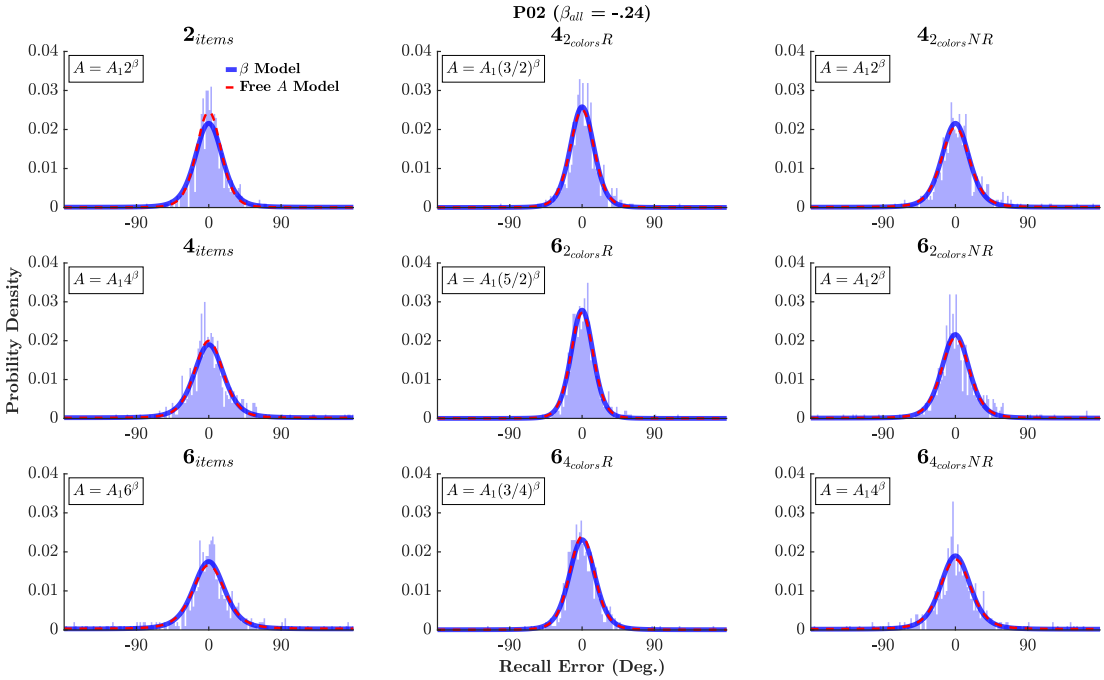


Figure C1. P02 empirical histograms overlaid by a 1β model with probability summation (blue-line) and free amplitude model fits (red-dash). Annotations denote conditional amplitude calculations. N is color count and all conditions follow a shared β power-law, with redundant elements benefiting from probability summation.

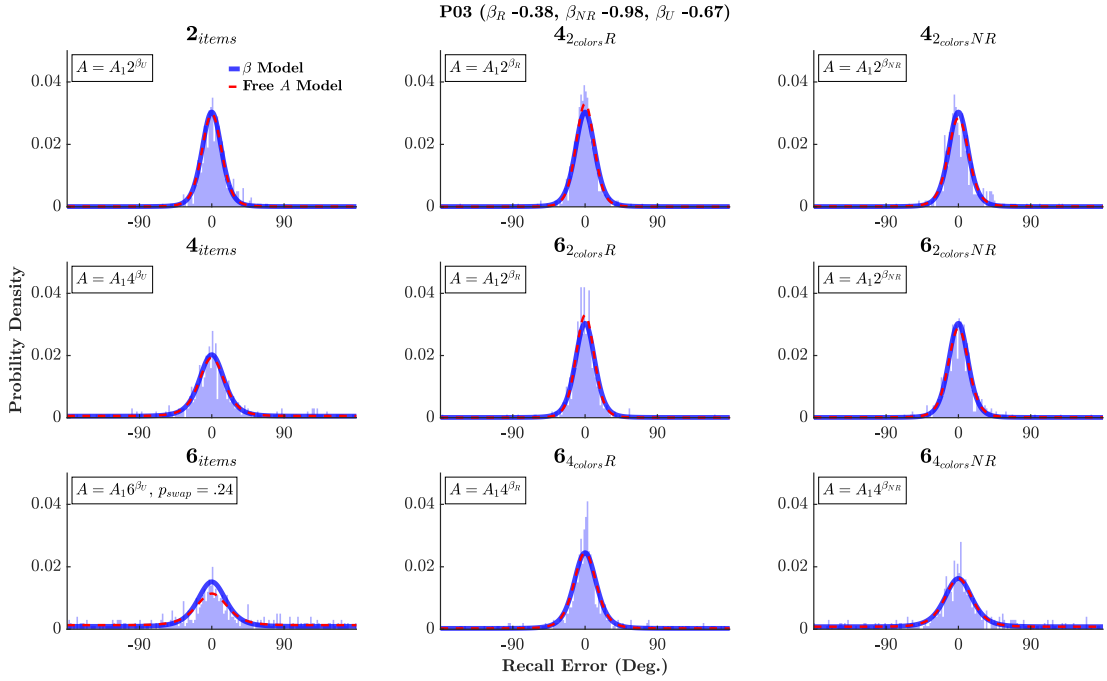


Figure C2. P03 empirical histograms overlaid by a 3β model (blue-line) and free amplitude model fits (red-dash). Annotations denote conditional amplitude calculations. N is color count. Redundant elements did not benefit from probability summation.

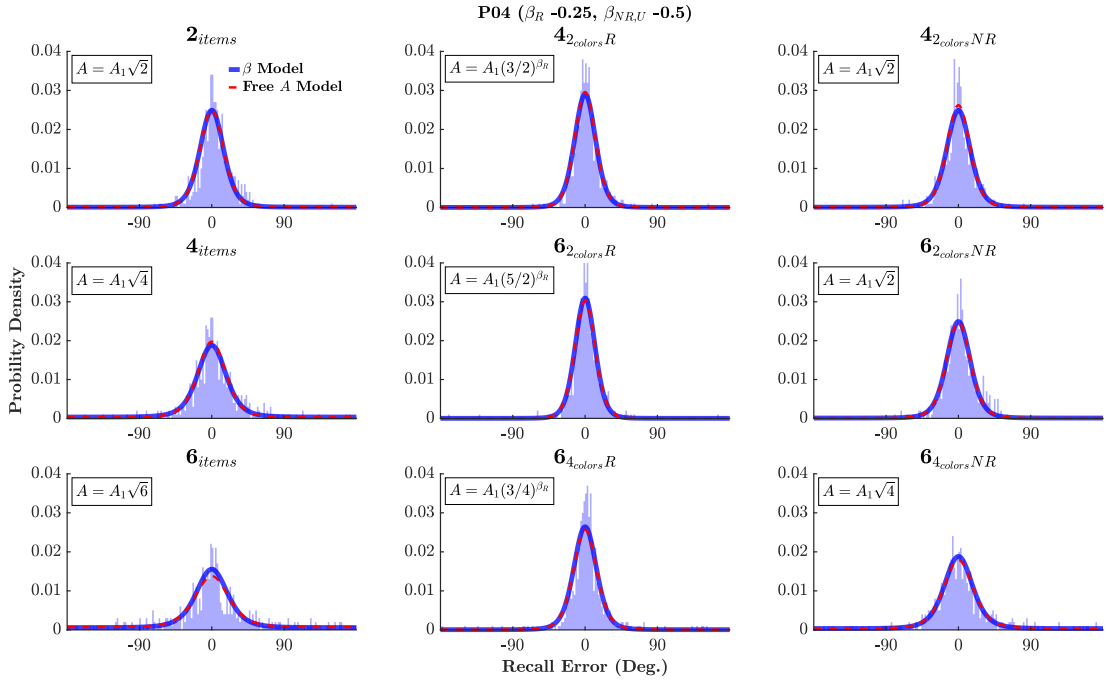


Figure C3. P04 empirical histograms overlaid by a 1β model with probability summation (blue-line) and free amplitude model fits (red-dash). Annotations denote conditional amplitude calculations. N is color count, redundant items follow a β_R power-law with probability summation, and non-redundant and unique arrays follow a square-root sample-size relationship.

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